

Panayiota Papafiggou

MSc Computer Science Student | Artificial Intelligence & Machine Learning

United Kingdom | +44 7763 842381 | yiotapapafiggou@gmail.com

LinkedIn | Personal Website

Profile

MSc Computer Science student at the University of Warwick with research interests in artificial intelligence, reinforcement learning, multi-agent systems, and intelligent decision-making. Experienced in computational modelling, simulation, data analysis, and formal methods, with a long-term interest in academic research and teaching.

Education

University of Warwick

MSc Computer Science

Sep 2025 – Present

Coventry, UK

- Relevant study: Machine Learning, Data Analytics, Natural Language Processing, and advanced Computer Science.

University of Nottingham

BSc (Hons) Computer Science

Sep 2022 – Jun 2025

Nottingham, UK

- Dissertation: *A Study of Synthetic Differential Geometry in Type Theory*.
- Relevant study: Artificial Intelligence, Software Engineering, Computer Security, Human-AI Interaction, and Formal Methods.

Petrou & Pavlou Lyceum

Apolytirion: 19.7/20

Sep 2019 – Jun 2022

Limassol, Cyprus

- A Levels: Pure Mathematics (A) and English (A).

Research Experience

MSc Dissertation Research

University of Warwick

Sep 2025 – Present

Coventry, UK

Assortative Matching and the Emergence of Cooperation in Multi-Agent Reinforcement Learning Systems

- Investigating how assortative matching and partner selection influence cooperation among reinforcement learning agents.
- Developing multi-agent models in Python using Q-learning, partner switching, and assortative rematching.
- Conducting sensitivity analyses, ablation studies, population experiments, convergence analysis, and behavioural analysis.
- Evaluating cooperation, learned strategies, and agent behaviour through large-scale simulation and statistical analysis.

External Research Collaborator

Cyprus University of Technology

Jun 2025 – Aug 2025

Limassol, Cyprus

- Developed vessel-movement detection algorithms using Automatic Identification System (AIS) maritime data.
- Analysed noisy and incomplete observations, reporting errors, and AIS message limitations.
- Contributed to maritime monitoring research under the supervision of Associate Professor Herodotos Herodotou.

Undergraduate Dissertation Research

University of Nottingham

Sep 2024 – Jun 2025

Nottingham, UK

A Study of Synthetic Differential Geometry in Type Theory

- Formalised concepts from Synthetic Differential Geometry using Homotopy Type Theory and Agda.
- Developed mechanically verified definitions and proofs involving curvature, metrics, and geometric structures.
- Produced formal implementations and mathematical visualisations.

Industry Experience

Software Engineer Intern <i>Amdocs</i>	Jun 2024 – Jul 2024 <i>Limassol, Cyprus</i>
- Supported software testing and automation using Unix, Jenkins, Postman, Octane, and Ginger by Amdocs. - Worked with CI/CD pipelines, API testing, debugging, verification, and validation.	
Summer Intern <i>Golden Union Shipping</i>	Jun 2023 – Sep 2023 <i>Limassol, Cyprus</i>
- Supported migration to a cloud-based crew-management platform through process analysis and data verification. - Assisted with system transition, data quality, and regulatory-compliance activities.	

Selected Technical Projects

AI for Multi-Agent Coordination in Space Exploration	
Developed a 3D rover simulation in Python and compared PPO, MADDPG, and evolutionary algorithms under navigation, energy, and coordination constraints.	
NLP-Based Travel Booking Chatbot	
Developed a Python conversational AI system using intent recognition, information retrieval, and sentiment analysis.	
Nottingham & Union Rowing Club Application	
Co-developed a mobile application to improve navigation, accessibility, and member interaction.	

Technical Skills

Programming	Python, Java, C, C++, JavaScript, PHP, SQL, HTML, CSS
AI & Machine Learning	Reinforcement Learning, Q-Learning, Multi-Agent Systems, PPO, MADDPG, Evolutionary Algorithms, NLP
Research Methods	Computational Modelling, Experimental Design, Sensitivity Analysis, Ablation Studies, Statistical Analysis, Simulation
Scientific Computing	NumPy, pandas, Numba, Matplotlib, PettingZoo, Ursina
Formal Methods	Agda, Homotopy Type Theory, Dependent Type Theory, Formal Proofs, LaTeX
Software & Tools	Git, GitHub, Linux, Unix, Jenkins, CI/CD, Postman, MySQL, Agile, Scrum

Leadership & Academic Activities

Computing Officer	University of Nottingham MedTech; supported technical activities, events, and workshops.
Hackathon Organiser	Co-organised an interdisciplinary MedTech hackathon.
European Youth Parliament	Experience in public speaking, teamwork, leadership, and debate.
Academic Societies	Warwick AI Society, Warwick Computing Society, Coding Society, and HackSoc.
Sport & Music	AEL Volleyball Team, Cyprus; Classical Guitar, Grade 5.

Languages

Greek: Native	English: Fluent
----------------------	------------------------